# Caesar Cipher:- IS - Lab-2

**Q1** Plain Text = ATTACK

Key = 5

Plain Number = 0 19 19 0 2 10

Cipher = (P+K) % 26

C ⇒ A=0 → 0+5 = 5 → (F)

T=19 → 19+5 = 24 → (Y)

C=2 → 2+5 = 7 → (H)

K=10 → 10+5 = 15 → (P)

**C = FYYFHP**

**Q2** Cipher Text = WKLVLVIXQ

Key = 3

Plain Text ⇒ W=22 → 22-3 = 19 → (T)

K=10 → 10-3 = 7 → (H)

L=11 → 11-3 = 8 → (I)

V=21 → 21-3 = 18 → (S)

I=8 → 8-3 = 5 → (F)

X=23 → 23-3 = 20 → (U)

Q=16 → 16-3 = 13 → (N)

Plain Text = **THIS IS FUN**

**Q3** P=XYZ , Key=4

C ⇒ X=23 → 23+4 = 27 % 26 = 1 → (B)

Y=24 → 24+4 = 28 % 26 = 2 → (C)

Z=25 → 25+5 = 29 % 26 = 3 → (D)

Cipher = **BCD**

0-A
1-B
2-C
3-D
4-E
5-F
6-G
7-H
8-I
9-J
10-K
11-L
12-M
13-N
14-O
15-P
16-Q
17-R
18-S
19-T
20-U
21-V
22-W
23-X
24-Y
25-Z

CamScanner

Q4 ⇒ Answer:-

There are 25 effective keys in a standard English Caesar Cipher.

It is easy to break because:

- No complex algorithm required, hacker just needs to shift using only 25 keys, (brute force).
- All characters are shifted by just 1 key, so its easy to decrypt the entire message.

⇒ Vigenère Cipher :-

Q1 P = ATTACK , Key word = KEY

K = KEYKEY

Encryption:

K+A = K+0 → (K)

~~E+T → 4+19 = 23 → (X)~~

E+T → 4+19 = 23 → (X)

~~Y+T → 24+19 = 43-26 = 17 → (R)~~

K+Y → 10+24 = 34-26 = 8 → (I)

E+C → 4+2 = 6 → (G)

Cipher = KXRKGI

Q2 P = SUNRISE

K = CATCATC

Encryption:

S+C → 18+2 = 20 → (U)

U+A → (U)

(N)+T ⇒ 13+19 = +~~52~~26 = ~~26~~ → (G)

(R)+C → 17+2 = 19 → (P)

(I)+A → 8+0 = 8 → (I)

(S)+T → 18+19 = 37+26 = ~~11~~ → (L)

(E)+C → 4+2 = 6 → (G)

Cipher = UUGTTILG

Q3 C = UUGTTILG

K = CATCATC

Decryption: (U)-C → 20-2 = 18 → (S)

(U)-A → 20-0 = 20 → (U)

(G)-T → 6-19 = -13+26 = 13 → (N)

(T)-C → 19-2 = 17 → (R)

(I)-A → 8-0 = 8 → (I)

(L)-T → 11-19 = -8+26 → (S)

(G)-C → 6-2 = 4 → (E)

P = SUNRISE

Q4

Ans:- Because both 'L's were added with different ~~Re~~ chars, as the keyword repeats itself for spanning the whole plain text, and it also has multiple characters.

## ⇒ Auto-Key Cipher :-

**Q1** P = SECRET
K = KEYSEC

Encryption :-

(S) + K → 18 + 10 = 28 - 26 = 2 → (C)
(E) + E → 4 + 4 = 8 → (I)
(C) + Y → 2 + 24 = 26 - 26 = 0 → (A)
(R) + S → 17 + 18 = 35 - 26 = 9 → (J)
(E) + E → 4 + 4 = 8 → (I)
(T) + C → 19 + 2 = 21 → (V)

C = CIAJIV

**Q2** C = FEGHRIQ
K = CAB

Decryption :-

(F) - C → 5 - 2 = 3 → (D)
(E) - A → 4 - 0 = 4 → (E)
(G) - B → 6 - 1 = 5 → (F)
(H) - D → 7 - 3 = 4 → (E)
(R) - E → 17 - 4 = 13 → (N)
(I) - F → 8 - 5 = 3 → (D)
(I) - E → 8 - 4 = 4 → (E)
(Q) - N → 16 - 13 = 3 → (D)

P = DEFENDED

CamScanner

Q3

Ans:- 1 wrong letter can corrupt every later after it because this key is built from plain text.

⇒ Play-Fair Cipher:-

Q2 P = BALLOON , fillingCharacter = X

Playfair Pairs = BA LX LO ON

Q1 Keyword = SECURITY

Grid =

| | | | | |
|---|---|---|---|---|
| S | E | C | U | R |
| I/J | T | Y | A | B |
| D | F | G | H | K |
| L | M | N | O | P |
| Q | ~~V~~ | W | ~~X~~ | Z |

Q3 Keyword = MONARCHY
PlainText = HOMEMO

- HO → FH
  → Same Column
- ME → CL
  → Same Column
- MO → ON
  → Same Row

Grid =>

| | | | | |
|---|---|---|---|---|
| M | O | N | A | R |
| C | H | Y | B | D |
| E | F | G | I/J | K |
| L | P | Q | S | T |
| U | V | W | X | Z |

Pairs = HO ME MO

C = FH CL ON

Q4 Keyword = MONARCHY

Cipher = CFSUPM

Grid ⇒

| | | | | |
|---|---|---|---|---|
| M | O | N | A | R |
| C | H | Y | B | D |
| E | F | G | I/J | K |
| L | P | Q | S | T |
| U | V | W | X | Z |

Cipher Pair = CF SU PM

- CF → HE
- SU → LX
- PM → LO

Plain Playfair pairs = HE LX LO

**PlainText = HELLO**

Q5

Ans:- Two identical letters cannot come in the same pair because the encrytion rules (same row, col— or squre/rectangle) can't be implemented on same letter. So, we add a filled letter like 'X'.

# IS-Lab02 CP-Tasks

## Approach 1:

Keyword = ~~SECURE~~ SERVICE

Key = 3

Caesar-Ciphered Keyword = VHUYLFH

Name = HUSSAIN

Pair = HU SX SA IN

Grid ⇒

| | | | | |
|---|---|---|---|---|
| V | H | U | Y | L |
| F | A | B | C | D |
| E | G | I/J | K | M |
| N | O | P | Q | R |
| S | T | W | X | Z |

→ HU → UY

→ SX → TZ

→ SA → TF

→ IN → EP

Encrypted Name = UYTZTFEP